

Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	3 (Term 1)	**Year:**	6
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to explain how sensors, a controller and actuators work together in a robot. They will also learn to trace a feedback loop and explain how a control system responds to changing input. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Ask pupils to name each component and explain the chain when an obstacle is detected. Check the sensor-to-controller-to-output path and that pupils include repeated feedback, not a one-time check. Note pupils who need further guided practice.

LESSON SLOTS - WEDNESDAY AND THURSDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	9 September 2026	By the end of this lesson, students will be able to explain how sensors, a controller and actuators work together in a robot.	**H:** Ask how a robot cleaner could notice an obstacle. **ID:** Introduce sensors as inputs, a controller as the instruction-processing part and actuators as parts that create movement. **WD:** Trace obstacle detected, controller decides to turn, motors turn the wheels; use arrows to show information and action. **YD:** Pupils label a robot diagram and act out sensor, controller and actuator roles for two situations. **C:** Ask pupils to name each component and explain the chain when an obstacle is detected.	**Support:** Provide labelled robot diagrams, vocabulary prompts and control-loop cards. Accept a spoken explanation before writing; rehearse one example together. **Extension:** Ask pupils to test an unusual input, explain a possible failure and improve the design without changing its main purpose.
2	10 September 2026	By the end of this lesson, students will be able to trace a feedback loop and explain how a control system responds to changing input.	**H:** Ask how an air conditioner knows when to stop cooling. **ID:** Explain repeated sensing, comparison with	**Support:** Provide labelled robot diagrams, vocabulary prompts and control-loop cards. Accept a spoken explanation before writing; rehearse one example

			a target and switching an output on or off. **WD:** Use temperature cards to model sensing above a target, switching cooling on, sensing again and stopping at the target. **YD:** Pairs arrange a control-loop diagram and test it against warmer and cooler temperature cards. **C:** Check the sensor-to-controller-to-output path and that pupils include repeated feedback, not a one-time check.	together. **Extension:** Ask pupils to test an unusual input, explain a possible failure and improve the design without changing its main purpose.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Primary Computing Student Book 6 printed pages 10-13
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Prepared robot pictures, labelled component diagrams and control-loop cards
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Prepared examples and the lesson success checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?